

Vincent T. Cook

ROBOTICS & MECHATRONICS

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Aerospace engineering senior with robotics and EV powertrain design experience. Apply CAD, structural analysis, MATLAB/Simulink, and dynamics modeling to electromechanical assemblies and prototype development.

EDUCATION

San Jose State University | San Jose, CA
B.S. Aerospace Engineering | Expected May 2027 | Enrolled Sep 2022

TECHNICAL SKILLS

Design & analysis: SolidWorks, Autodesk Inventor, CAD, stress analysis, simulation and modeling, CFD, thermal testing
Programming & dynamics: MATLAB/Simulink, MotionGenesis, Python, C++, Java | Electromechanical integration

ENGINEERING EXPERIENCE & PROJECTS

SJSU Robotics

Mechanical Designer | Aug 2024–Present

- Designed a radio transmitter mast incorporating a camera mount and sensors for a student robot.
- Refined robot designs using stress analysis to balance weight and structural integrity; reinforced high-stress regions and used lightweight materials in lower-load sections.

SAE Spartan Racing

EV Powertrain Intern | Aug 2023–May 2025

- Created 3D CAD models and assembly instructions for an electric motor performance test bench integrating electrical and mechanical components.
- Worked with teammates to resolve technical challenges and coordinate engineering and manufacturing requirements.

Hybrid Electric / Hydrogen Fuel-Cell Aircraft | SJSU Senior Project

Aerospace Engineering Team Member — In Progress | 2026–Present

- Contribute to wing-model design and planning for a yearlong hybrid battery-electric and hydrogen fuel-cell model aircraft project.
- Planned work includes CAD, CFD and thermodynamic analysis, aircraft construction/testing, and battery/fuel-system electronics integration.
- Planned scope also includes commercial-scale modeling; the team targets first flight in spring 2027.

BROADSWORD | Reusable Rocket Concept

CAD & Programming Subteam Lead | Sep–Dec 2023

- Developed landing-leg CAD models and simulation code generating trajectory-angle-versus-time plots for a reusable rocket concept.
- Coordinated CAD and programming subteams, delegated tasks, and provided design feedback.

NASA L'SPACE Mission Concept Academy

Mechanical Engineering Team Member — Educational Program | Fall 2026–Present

- Help plan scientific approaches and evaluate mechanical-component options for a robotic lunar surface payload mission concept.

SELECTED COURSEWORK

- AE 138 / AE 30 | Flight & trajectory modeling: Used MATLAB and MotionGenesis to model 3D rigid-body Frisbee flight from initial to final position; coded missile launch trajectories.
- AE 157 | Control systems: Applied MATLAB and Simulink to control-system coursework.
- AE 20 | Motor CAD: Created high-level motor designs in introductory CAD coursework.

WORK EXPERIENCE

The Tech Interactive | Educator I | 2024–2025

- Taught hands-on STEM labs and design challenges, explaining technical concepts to diverse learners.

Self-Employed | Private Swim Instructor | 2023

- Managed client scheduling and developed individualized lessons based on skills and goals.

YMCA & Ventura Aquatics Center | Swim Instructor / Coach / Lifeguard | 2021–2024

- Coached swimmers, led practices, and adapted instruction for children with disabilities.